

**The Effect of *Pleurotus ostreatus* in Bioremediation of *Escherichia coli* Contaminated
Water Grown on Various Substrates**

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Abstract

Bacterial contaminant in water is a major issue that causes many deaths every year (Pini & Geddes, 2020). The type of substrate the fungi grows on can greatly affect which nutrients the fungi receives which in turn affect the enzymes that it makes (El-Ramady et al., 2022). Many different things in a substrate affect the remediation rate of the fungi, such as the carbon to nitrogen ratio (El-Ramady et al., 2022). The purpose of this study was to investigate which substrate coffee, sawdust, or wheat straws, would have the highest remediation capability. It was hypothesized that if *Pleurotus ostreatus* grown on wheat straws, would inhibit the growth of *E. coli* the greatest. Compared to coffee grounds and sawdust because wheat straws have the highest percentage composition of carbohydrates meaning that they would provide the most nutrients for the fungi to grow (Gashaw et al., 2020). The fungi was grown with the substrate for three weeks, before the *E. coli* broth was added. After 15 days, the bacterial growth was quantitated using a colony counter. Coffee and sawdust substrate was found to have the least number of colonies than wheat straw substrate; the one-way ANOVA test shows that the data was significant $F(3,116) = 45.28$, $p < .0001$. These findings did not support the hypothesis but rather support that sawdust and coffee provide *P. ostreatus* improved nutrition to inhibit *E. coli* growth. Therefore sawdust and coffee would be better substrates for *P. ostreatus* to remediate *E. coli* than wheat straw.

Introduction

Pollution has been affecting the earth in recent years. An influx of bacterial contaminants in the air, water, and land have caused various types of cancer (Pini & Geddes, 2020). But many biotic contaminants in the form of bacteria *Shigella spp.*, *Giardia lamblia*, and *Escherichia coli* also cause major health issues and environmental damage (Pini & Geddes, 2020). Bacteria can easily pollute waterways such as rivers and lakes. Rivers are vital to development of urban environments (Zhang et al., 2020). Development has caused industrial wastewater to be dumped into rivers which pollutes aquatic environments (Zhang et al., 2020).

Escherichia coli is a bacterium species that has many different uses. Some *E. coli* can be beneficial for example the *E. coli* found in the gastrointestinal system. But *E. coli* can also be pathogenic which can affect any part of the body and kill. In America alone in the year 2000, 40,000 people died from blood infection due to *E. coli* which is an extremely hazardous bacterium due to antibiotic and infection resistance (Pini & Geddes, 2020). *E. coli* is used often in research because of its pathogenic and degradative enzymes similar to what fungi produce (Pini & Geddes, 2020). The concentration of *E. coli* can be determined using the number of colonies.

Current treatment methods are often expensive, ineffective, or take too long. Current treatments include physical, chemical, mechanical, biological methods. Many conventional water treatments create sludge byproducts which need to be disposed of and might need pre or post treatment (Crini & Lichtfouse, 2018). Chemical precipitation is an example of conventional water treatment but it creates sludge which needs to be disposed of (Crini & Lichtfouse, 2018). Biological methods are good treatment methods depending on the species of microorganism used (Crini & Lichtfouse, 2018). Some of them need favorable environments and are a slow process

(Crini & Lichtfouse, 2018). Electrochemistry water treatments have a high initial and maintenance cost and require a post treatment (Crini & Lichtfouse, 2018). Membrane filtration and reverse osmosis are good treatment options because it produces high quality treatment but it requires high energy and maintenance and also has a limited output rate (Crini & Lichtfouse, 2018).

Green biotechnology, which is the use of genetic engineering and tissue culture, is the key to solving this issue; often referred to as red biotechnology in the medical field and white biotechnology when used in the industrial field (El-Ramady et al., 2022). Green biotechnology can engineer organisms to remediate, which means to get rid or degrade, harmful pollutants and organisms from the environment. Mycoremediation involves using fungal cultures to remediate pollutants; the mycelium roots release digestive enzymes that break down pollutants (El-Ramady et al., 2022). The remediation capabilities of fungi are efficient on many different entities, such as antibiotics, bacteria, dyes, and plastic (Jayasinghe et al., 2008, Singh et al., 2017, Taylor et al., 2015, Matavuly & Molitoris, 2009). Remediation results in products that are less toxic to the environment and fungal cultures can be used to remediate land and marine pollutants (El-Ramady et al., 2022; Matavuly & Molitoris, 2009; Jayasinghe et al., 2008).

Species of white-rot fungi, especially *Phanerochaete chrysosporium*, *Pleurotus ostreatus*, and *Trametes versicolor*, have been used in research because of their properties, such as non-specific growing conditions, rate of growth, and degradation of pollutants (El-Ramady et al., 2022). These fungi have been used as antibiotics, antitumors, antivirals, etc., indicating their vast medicinal potential (Gashaw et al., 2020). *P. ostreatus*, commonly called oyster mushroom, is an edible mushroom, often identified by the shape of its fruiting bodies (Singh et al., 2017). They can be grown on a variety of agricultural products at temperatures between 15°C to 30°C

and have been used in many past studies involving mycoremediation (Gashaw et al., 2020). *P. ostreatus* has high remediation properties due to its enzymes use of laccase, manganese peroxide, lignin peroxidases, xylanases, etc. (Singh et al., 2017). Allowing for the fungi to effectively use the substrate it has grown on and degrade pollutants which is also called mycoremediation. The enzymes released by *P. ostreatus* are called ligninolytic enzymes, which have high degradation properties (El-Ramady et al., 2022). These degradation properties often depend on what nutrients the fungi has available to use.

The fungi are cultivated on various substrates; characteristics of substrates, like crude proteins, carbohydrates, fats, fiber, ash, etc., can significantly impact the bioremediation properties of the fungi (El-Ramady et al., 2022). Many agro-industrial products such as wheat straws, barley straws, rice straws, maize cob, soya stalk, cotton stalk, cotton seed hull, rice husk, sugarcane bagasse, cassava peel, acacia sawdust, beech sawdust, birch sawdust, coffee grounds are used as substrates (El-Ramady et al., 2022). The carbon to nitrogen ratio in substrates influences how the fungi performs in the degradation rates (El-Ramady et al., 2022). Because that is going to affect the protein content, which in turn affects the enzymes made by the fungi (El-Ramady et al., 2022).

Many fungal species have high remediation properties, such white-rot fungi, which can be altered because of the substrate they are grown on. In recent years, there has been an increase in the usage of synthetic antibiotics which has led researchers to develop a plant based antimicrobial to aid in this issue since synthetic antibiotics have environmental damage (Gashaw et al., 2020). *P. ostreatus* was a viable organism which could be used since it was also used in previous studies to degrade various pollutants and organisms such as bacteria and antibiotics. Gashaw et al. (2020) researched how *P. ostreatus* antibacterial properties can change when

grown on different substrates. The substrates used were coffee straw, sorghum grain residue, pea straw, and wheat grain (Gashaw et al., 2020). Five bacterial species were used to test antibacterial properties; *E. coli*, *B. subtilis*, *S. faecalis*, *P. aeruginosa*, and *S. typhi* (Gashaw et al., 2020). The research shows that methanolic extracts, which are the extracts of antibacterial compounds, have the highest inhibitory effect on *E. coli*, showing that *P. ostreatus* is very effective at stopping and killing *E. coli* (Gashaw et al., 2020). The results of the study show how the different substrates used produce different remediation rates of the bacteria depending on carbon to nitrogen ratio and chemical composition. A study done by Singh et al. (2017) uses *P. ostreatus* to degrade ciprofloxacin (CIP), an antibiotic. The study found that *P. ostreatus* was successful at degrading the CIP (Singh et al., 2017). A study done by Pini and Geddes (2020), used *P. ostreatus* mycelium grown on a substrate to remove *E. coli* from water to demonstrate that it can help improve river contaminated water. The study shows that *P. ostreatus* was able to take *E. coli* out of the water (Pini & Geddes, 2020). All three of these studies show how effective *P. ostreatus* is at remediation and how it can be used to remediate various types of bacteria and other pollutants.

The purpose of this experiment was to see how *P. ostreatus* grown on different substrates could affect the rate of growth of *E. coli*. The substrates that were used included wheat straws, coffee grounds, and sawdust, because these substrates are commonly used when growing fungi. Furthermore, there are no previous studies comparing these three substrates and their effects on *E. coli*. It was hypothesized that if *Pleurotus ostreatus* were grown on wheat straws, then *P. ostreatus* would result in a decreased survival rate of *E. coli*. Compared to coffee grounds and sawdust because wheat straws have the highest percentage composition of carbohydrates meaning that they would provide the most nutrients for the fungi to grow (Gashaw et al., 2020).

The growth of *E. coli* was tested by counting the number of colonies using a colony counter. At the end of the experiment the number of colonies were counted and compared with the other experimental groups to determine which substrate had the greatest inhibitory effect on *E. coli* growth.

Materials

Pleurotus ostreatus Culture Plate (Out Grow)	Escherichia coli K-12 Nutrient Broth (Carolina Biological)
Wheat Straws (Amazon)	Petri Dish
Coffee Grounds (Amazon)	Lennon LB Agar Powder
Pine Sawdust (Amazon)	Colony Counter
Deli Cups	Bleach
Distilled Water	Metal Straw
Dextrose	Pipette
Micropipette 25µL, 1000 µL	Stereoscope
Inoculation Loop	Aluminum Foil
Permanent Marker	Red Potato
Incubator	Scoopula
Autoclave	Beaker 10 mL, 500 mL
Muslin Cloth	Ethyl Alcohol
Stirring Rod	Erlenmeyer Flasks 50 mL, 100 mL, 500 mL
Graduated Cylinder 10 mL, 100 mL, 200 mL, 1000 mL	

Methodology

The substrates were first prepared by sterilizing and weighing them. Then the fungi was added for it to grow on the substrate. The *E. coli* broth was made and then added to the cups. The *E. coli* was then taken out and quantified.

All lab equipment during the experimentation process was disinfected with ethyl alcohol before use. First the wheat straws were cut into 3 to 5 centimeter pieces and put into a 500 mL beakers and covered with aluminum foil. Similarly coffee grounds and pine sawdust were put into separate beakers. Then the substrate beakers were autoclaved to sterilize the substrate. The

substrates were massed 2.0 ± 0.1 grams of substrate in each cup using a scoopula and a scale. Using this process 30 cups of each substrate were prepared. The deli cups were numbered by substrate from 1 to 120 starting with 1-30 for coffee, 31-60 for sawdust, 61-90 for wheat straw, and 91-120 for control.

Han's (2021) research method was used to culture *Pleurotus ostreatus* growing on agar was obtained, and then plug-shaped holes were punched from the plate at about 1 cm in size. One plug was placed into each deli cup (120 plugs total) . The deli cups were covered to allow the fungi mycelium to grow. The cups were misted twice a week using a spray bottle to keep the mycelia moist and kept at room temperature. The fungi were allowed to grow for 3 weeks before adding the *E. coli* broth.

375 milliliters of dextrose broth were prepared according to Han (2021) using fresh potatoes, dextrose, and distilled water. First, one potato was peeled and cut into approximately 1 to 2 cm pieces then 50 g of potatoes were boiled 375 mL of distilled water in a beaker and filtered through a muslin cloth to obtain the extracts. Then 5 g of dextrose was mixed into the extract. The broth was autoclaved at 121°C for 30 min. 1260 mL of distilled water was then inoculated with 6 mL of K-12 *E. coli* nutrient broth from Carolina Biological. 375 mL of potato dextrose broth were added into the inoculated mix and stirred using a stirring rod. The *E. coli* solution was then incubated at 37°C for 48 h. After the solution was incubated for 48 hours, 10 mL of *E. coli* solution was measured using a graduated cylinder and added into each of 120 deli cups. After the *E. coli* solution was added, the cups were placed at room temperature for 15 days. Afterwards, a serial dilution was performed on each sample to find the concentration of *E. coli*. Lennon LB agar was prepared and autoclaved in accordance with the manufacturer instructions; 775 mL of the agar was prepared and poured into petri dishes after cooling the dishes and

refrigerated. A quadrants was drawn on each of the petri dishes using a permanent marker and a ruler.

The serial dilution was done by measuring 9 mL of water in three different test tubes. Then, using a 1 mL micropipette, pipette 1 mL the broth in the cup into the first test tube. After swirling the first test tube to mix the content pipette 1 mL of the first test tube into the second test tube. After swirling the second test tube pipette 1mL into the third test tube and swirling it. This made the factor of the serial dilution 1:1000. Then 25 μ L was pipetted onto one quadrant of a petri dish using a micropipette. Next an inoculation loop was used to spread the solution over the quadrant. The inoculation loop was sterilized between each trial using a bunsen burner. by holding the inoculation loop over the flame for a couple of seconds and then let it cool for a couple of seconds. Then the petri dishes were put in the incubator for 24 hours. Afterwards, they were taken out and the bacteria was quantified using a colony counter and a stereoscope. The colony counter was used to calculate the average number of colonies in each quadrant. The colony counter counted five squares on the grid, each square was 1 cm² and then the average of the five squares multiplied by the area of the quadrant equals the average number of colonies. An one-way ANOVA and a Tukey post hoc test were done to analyze the data. At the end of experimentation, all the living organisms used were placed in a 10% bleach solution to kill any living organism and were discarded.

Figure 1

Experimental Design Diagram

Experiment Title: The Effect of <i>Pleurotus ostreatus</i> in Bioremediation of <i>Escherichia coli</i> Contaminated Water Grown on Various Substrates				
Hypothesis: <i>Pleurotus ostreatus</i> grown on wheat straws, would inhibit the growth of <i>E. coli</i> the greatest. Compared to coffee grounds and sawdust because wheat straws have the highest percentage composition of carbohydrates meaning that they would provide the most nutrients for the fungi to grow (Gashaw et al., 2020).				
Independent Variable Type of substrate				
	<i>E. coli</i> (Control, no substrates)	Wheat straws	Coffee grounds	Sawdust
Number of Trials	30	30	30	30
Dependent Variable: Concentration of <i>E. coli</i> (number of colonies)				
Constants Incubation time, the type of fungi used, type of bacteria used, the incubation temperature, the fungal medium				
Control Trials with just <i>P. ostreatus</i> growing on agar				

Results

The bacteria was grown on agar petri dish for 24 h then the bacterial colonies counted.

The formula of the colony counter was used to find the number of colonies per trial. The 5 squares were averaged as seen in Table A1 then they were multiplied by the area to get the number of colonies as seen in Table A2. The means of the coffee (M=721.51) and sawdust (M=781.88) groups were much lower than the wheat straws (M=1149.88) and control group (M=1174.74). The control group and wheat straws have a higher variance, range, and standard deviation than coffee and sawdust.

Table 1*Descriptive Statistic of Coffee, Control, Sawdust, Wheat Straw*

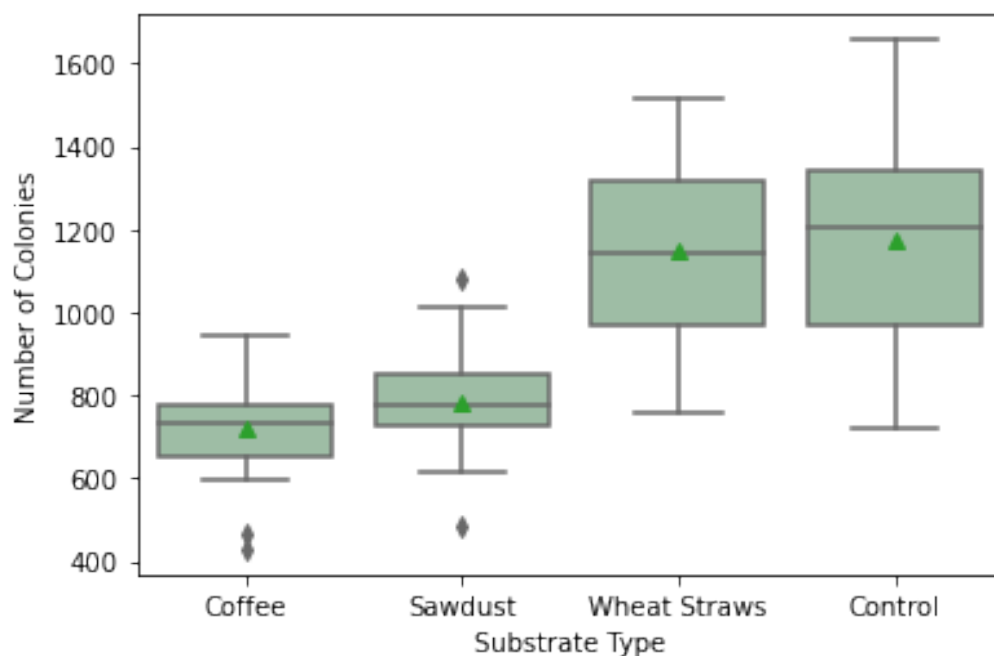
	Count	Mean	Standard Deviation	Minimum	Median	Maximum	IQR
Substrate Type							
Coffee	30	721.51	114.48	428.68	732.07	944.90	124.53
Control	30	1174.74	269.15	718.49	1206.03	1657.35	374.34
Sawdust	30	781.88	120.76	486.04	775.85	1077.73	128.30
Wheat Straw	30	1149.88	225.18	754.72	1141.13	1515.47	348.68

Note. Table 1 shows the mean, range, standard deviation, and variance for all the experimental groups.

Table 1 shows that the Coffee (M=721.5) and sawdust (M=781.9) groups were very similar in average number of colonies while wheat straws (M=1149.9) and control (M=1174.7) groups are much higher on average than sawdust and coffee. Two groups (Coffee and Sawdust) have a much lower mean than the other two groups (wheat straws and control). The lowest average number of colonies was the coffee group with an average of 721.508 colonies while the highest was the control group with an average of 1174.739 colonies.

Figure 2

Boxplot of Experimental groups Showing the Average Number of Colonies



Note. Figure 2 is a box plot that shows the interquartile range, the outlier in each group, and the median relative to each other.

The X axis represents the experimental groups. The Y axis represents the number of colonies.

Figure 2 shows the distribution of each group. Coffee and sawdust groups are placed much lower than wheat straw and control groups which mean the distribution of coffee and sawdust trials are lower when compared to control and wheat straws. The range of coffee and sawdust are also smaller than wheat straws and control.

Table 2

One-Way ANOVA Summary Table

ANOVA						
Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	5122094.97	3	1707364.99	45.28	<0.0001	2.68
Within Groups	4374343.24	116	37709.856			

Note. Table 2 shows the One-way ANOVA statistic that was run to analyze the data

The one-way ANOVA test shows that the data was significant $F(3,116) = 45.28$, $p < .0001$. Which means that it failed to reject the alternative hypothesis and failed the null hypothesis which means there was a significant difference between the control group to the experimental group.

Table 3

Tukey Post Hoc Test

Comparison	Absolute Difference	Critical Range	Mean Significantly Different
1 to 2	60.4	131.1	no
1 to 3	428.4	131.1	yes
1 to 4	453.2	131.1	yes
2 to 3	368.0	131.1	yes
2 to 4	392.9	131.1	yes
3 to 4	24.9	131.1	no

Note. Table 3 shows the results of the tukey post hoc test. The mean difference shows if the comparison between the experimental groups is significant. 1 is coffee, 2 is sawdust, 3 is wheat straws, and 4 is control.

The comparison between coffee ($M=721.51$, $SD=114.48$) and sawdust ($M=781.88$, $SD=269.15$) was not significant but coffee and sawdust compared to control and wheat straws were significant. Similarly wheat straw ($M=1149.88$, $SD=225.18$) and control ($M=1174.74$, $SD=269.15$) groups did not have a significant difference between them.

Discussion

The purpose of this research was to investigate the type of substrate that would result in the best remediation of *E. coli* by *P. ostreatus* fungi. It was hypothesized that if *Pleurotus ostreatus* grown on wheat straws, would inhibit the growth of *E. coli* the greatest. Compared to coffee grounds and sawdust because wheat straws have the highest percentage composition of carbohydrates meaning that they would provide the most nutrients for the fungi to grow (Gashaw et al., 2020). This hypothesis was not supported according to the data. The study found that when wheat straw, coffee, and sawdust used as substrates coffee and sawdust have a higher inhibition of *E. coli* growth when compared to wheat straw.

The one-way ANOVA test found that there was a statistically significant difference between the fungi with the substrates and the fungi without the substrate. According to Table 4 the Tukey test shows that that data between the coffee and sawdust was not statistically significant which means that they had a similar number of colonies meaning that both substrates performed the same when it came to inhibiting *E. coli* growth. The Tukey post hoc test results between the wheat straws and control groups were not statistically significant, which means they also had a similar number of colonies. But the Tukey test showed a significant difference between the coffee-sawdust and the wheat straw-control group meaning that some substrates were better at inhibiting *E. coli* growth than other substrates. The substrates that had the best results for remediation of *E. coli* were coffee and control. There was no statistical difference between control and wheat straw groups which might be because the fungi did not grow on the wheat straws. Although some of the trials did make it seem like there was visible growth of the fungi in those cups. Overall there was visible evidence of over growth of the *E. coli* on the agar petri dishes, which they were grown on for the wheat straws and control group. While there was

no overall visible overgrowth where the bacterial colonies grew on top of each other on the coffee and sawdust groups. The standard deviation, range, and variation from Table 2 for coffee and sawdust was very low, demonstrated the consistency of the trials.

A study done by Han (2021) showed that when *P. ostreatus* was grown on wheat straws, the *E. coli* concentration increases from the initial amount. Similarly, in this study, wheat straw was not the most effective substrate for the fungi to remediate the bacteria. Which demonstrates that the wheat straw does not provide *P. ostreatus* with the best nutritional medium for *E. coli* remediation. Although Han (2021) was not specifically investigating how different types of substrates affect remediation of *E. coli*, the study showed that wheat straws do not provide the best results with *P. ostreatus* compared to other types of fungi. A study by Gashaw et al. (2020) found that methanolic extract of *P. ostreatus* coffee straws produced the worst results of bioremediation of *E. coli* compared to the other substrates. The other substrates included *Sorghum* grain residue, pea straw (which showed the best results), and wheat grain (Gashaw et al., 2020). The current study found that coffee out performed wheat straw which differs from what Gashaw et al. (2020) found. The results might be because of the different methods and materials used. For example Gashaw et al. (2020) used methanolic extract, which are the extracts of antibacterial compounds of plants, of *P. ostreatus* while this study used *P. ostreatus* culture on growing on agar. Gashaw et al. (2020) were also comparing coffee straw with different substrates than this study and the method of quantifying bacterial growth was different. These differences might explain why different results were produced. A study done by El-Ramady et al. (2022) found that acacia, beech and birch sawdust have 45.2 % biological efficiency which is the ratio between fruiting bodies and substrate on *P. ostreatus* in comparison to 37.6% for wheat straw. Biological efficiency explains why the sawdust did better than the wheat straws. Also the

differences in the specific type of tree that the sawdust came from which might have caused the biological efficiency to differ. Pini & Geddes (2020) found that the *E. coli* concentration went down by 99.25% after 96 hours when *P. ostreatus* was grown on wheat straws. These results are different compared to what this study found since the *E. coli* concentration for wheat straws was similar to the control group. The study used different methods which might explain the difference in the results, for example, fewer trials, shaker table, and membrane filter. Taylor et al., (2015) found a 22% decrease in *E. coli* when *S. rugoso-annulata* was grown on wood chips which was the trial that had the greatest decrease in *E. coli*, which correlates with the results with sawdusts groups since both groups derive their nutrients from the wood in trees. The study was trying to observe the ability of *S. rugoso-annulata* to remove *E. coli* from storm water when grown on two different substrates (wood chips and a composition of wood chips and rice straws) (Taylor et al., 2015). Even though a different fungi was used, the study still showed the effectiveness of a wood derived substrate.

The results of this study showed that the best substrates to remediate *E. coli* from water were sawdust and coffee over wheat straws. This information helps determine the most effective substrate that *P. ostreatus* can be grown between the three most used and cost effective substrates used to grow fungi on and aids in making a sustainable solution of bacterial water contamination more efficient.

One factor that may have improved the reliability of the results was how much water the fungi received. The fungi were misted twice a week using a spray bottle; however this method did not ensure that all the trials got the same amount of water. This could be avoided by measuring out the water in a graduated cylinder and pouring it in. Another source of error could have been quantifying the bacterial colonies since some plates had overgrowth of bacteria where

the colonies were overlapping, making it difficult to accurately quantify the number of colonies. This source of uncertainty can be avoided by having a different method of quantitating the bacteria in the water. Furthermore, throughout the experiment, the cups were opened many times' increasing the chances of the fungi getting contaminated. To decrease the chances of contamination the cups could be opened in a sterile environment. Finally, the wheat straws experimental group cups did not have as much air as the other groups since the wheat straws had to be pushed into the cups to reach 2 g since the wheat straws had a lower density than the other substrates. Biggerer cups can be used therefore the wheat straws won't have to be pushed in. Also more time could be allowed for the fungi to grow and then more time for the fungi to remediate the bacteria.

Future studies may include examining how different types of substrates affect remediation of bacteria. For example, one potential study could look into how the coffee substrate can affect the remediation of *E. coli* compared to *Staphylococcus*. In addition, very little information is available on how the substrates affect different types of bacteria remediation when grown on a particular fungi, which could be a potential research study that would be of great use in this field. For example how would using *Phanerochaete chrysosporium* compared to *Trametes versicolor* affect the remediation of *E. coli* when grown on different substrates. One final future research study could focus on a mixture of the substrates and the various impacts for combinations of fungal remediation. For example, coffee grounds and sawdust could be mixed together to test for effects on remediation rates.

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References

- Crini, G., & Lichtfouse, E. (2018). Advantages and disadvantages of techniques used for wastewater treatment. *Environmental Chemistry Letters*, 17(1).
<https://doi.org/10.1007/s10311-018-0785-9>
- El-Ramady, H., Abdalla, N., Fawzy, Z., Badgar, K., Llanaj, X., Törös, G., & Hajdú, P. (2022). Green biotechnology of oyster mushroom (*Pleurotus ostreatus* L.): A sustainable strategy for myco-remediation and bio-fermentation. *Sustainability*, 14(6), 3667.
<http://dx.doi.org/10.3390/su14063667>
- Gashaw, G., Fassil, A., & Redi, F. (2020). Evaluation of the antibacterial activity of *Pleurotus* spp. cultivated on different agricultural wastes in Chiro, Ethiopia. *International Journal of Microbiology*, 2020, 9312489.
<https://doi.org/10.1155/2020/9312489>
- Han, M. (2021). The Mycoremediation of *Escherichia coli* by *Pleurotus ostreatus*, *Stropharia rugosoannulata*, and *Trametes versicolor* in contaminated water . *Journal of the South Carolina Academy of Science*, 19(2).
<https://doi.org/https://scholarcommons.sc.edu/cgi/viewcontent.cgi?article=1222&context=jscas>.
- Jayasinghe, C., Imtiaj, A., Lee, G. W., Im, K. H., Hur, H., Lee, M. W., Yang, H. S., & Lee, T. S. (2008). Degradation of three aromatic dyes by white rot fungi and the production of ligninolytic enzymes. *Mycobiology*, 36(2), 114–120.
<https://doi.org/10.4489/MYCO.2008.36.2.114>
- Matavulj, M., & Molitoris, H. (2009). Marine fungi: Degraders of poly-3-hydroxyalkanoate based plastic materials. *Zbornik Matice Srpske Za Prirodne Nauke*,

(116), 253–265. <https://doi.org/10.2298/zmspn0916253m>

Pini, A. K., & Geddes, P. (2020). Fungi are capable of mycoremediation of river water contaminated by *E. Coli*. *Water, Air, & Soil Pollution*, 231(2).

<https://doi.org/10.1007/s11270-020-4464-7>

Singh, S. K., Khajuria, R., & Kaur, L. (2017). Biodegradation of ciprofloxacin by white rot fungus *pleurotus ostreatus*. *3 Biotech*, 7(1). <https://doi.org/10.1007/s13205-017-0684-y>

Taylor A., Flatt A., Beutel., Wolff M., Brownson K., & Stamets P. (2015). Removal of *Escherichia coli* from synthetic stormwater using mycofiltration. *Ecological Engineering*, (78). <https://doi.org/10.1016/j.ecoleng.2014.05.016>.

Zhang, L., Tu, D., Li, X., Lu, W., & Li, J. (2020). Impact of long-term industrial contamination on the bacterial communities in Urban River sediments. *BMC Microbiology*, 20(1). <https://doi.org/10.1186/s12866-020-01937-x>

Appendix

Table A1

Raw data table with number of colonies counted for 5 squares using the colony counter and average

<https://docs.google.com/spreadsheets/d/>

[1JBwmrOYfqMeWRa1lXlO3SibB1CyXdXYrrlNLHQflkXg/edit?usp=sharing](https://docs.google.com/spreadsheets/d/1JBwmrOYfqMeWRa1lXlO3SibB1CyXdXYrrlNLHQflkXg/edit?usp=sharing)

Table A2

The average number of colonies for each experimental groups after the colony counter calculations

	Coffee	Sawdust	Wheat Straws	Control
1	464.9	624.9	1280.0	781.9
2	763.8	806.0	839.2	908.7
3	667.2	857.4	754.7	1594.0
4	685.3	950.9	806.0	1077.7
5	800.0	851.3	1274.0	1657.4
6	630.9	736.6	1328.3	987.2
7	887.5	612.8	1198.5	1322.3
8	703.4	860.4	1503.4	1328.3
9	649.1	790.9	1515.5	1174.3
10	944.9	748.7	1427.9	878.5
11	830.2	848.3	1153.2	1394.7
12	640.0	884.5	1304.1	1548.7
13	887.5	890.6	1482.3	1612.1
14	679.2	1077.7	1092.8	1210.6
15	781.9	724.5	1376.6	1346.4
16	733.6	760.8	1177.4	1207.5
17	730.6	682.3	914.7	851.3
18	751.7	670.2	1068.7	1228.7
19	594.7	763.8	1129.1	981.1
20	748.7	806.0	797.0	1041.5
21	609.8	486.0	1449.1	1213.6
22	800.0	787.9	1319.2	1277.0
23	724.5	854.3	1044.5	1400.8
24	727.5	624.9	938.9	718.5
25	742.6	1011.3	1307.2	1542.6
26	890.6	790.9	944.9	963.0
27	646.0	754.7	966.0	769.8
28	733.6	757.7	972.1	1204.5
29	766.8	727.5	1050.6	1159.2
30	428.7	712.5	1080.8	860.4